

AYAN BANERJEE

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EDUCATION

PhD Candidate, Computer Vision

Universitat Autònoma de Barcelona

📅 March, 2024 – Present

- Working on multi-object vector graphics synthesis with layout-guided diffusion models.

Master's in Computer Vision

Universitat Autònoma de Barcelona

📅 Oct, 2022 – Sep, 2023 💎 GPA: 8.08/10.0

- Thesis Title: Exploring graph-based knowledge distillation towards document layout analysis.

EXPERIENCE

Research Intern

Institute for People-Centred AI, University of Surrey

📅 Jan, 2025 – Feb, 2025

- Working on sketch-based comic character editing.

Research Intern

Computer Vision Center

📅 June, 2022 – Sep 2022

- Working on semi-supervised learning for document layout analysis.

Project Linked Person

Indian Statistical Institute, Kolkata

📅 Sep, 2021 – May, 2022

- Working on an effective deep learning algorithm towards end-to-end scene text spotting.

Data Science Intern

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📅 July, 2020 – August, 2020

- Exploratory data analysis for augmented reality to obtain virtual trials on shoes.

PUBLICATIONS

- Banerjee A**, Mathur N, Lladós J, Pal U, Dutta A. CraftSVG: Multi-object text-to-SVG synthesis via layout-guided diffusion (WACV 2026).
- Banerjee A**, Vilariño F, Lladós J. CraftGraffiti: Exploring Human Identity with Custom Graffiti Art via Facial-Preserving Diffusion Models (NeurIPS Creative AI 2025)

VOLUNTARY WORKS

- Serving as reviewer for top conferences (NeurIPS, ICLR, ICML, CVPR, BMVC).
- Supervised two Master's student projects on vector graphics editing and document restoration.

PROJECTS

Explorations of generative models for multi-object visual creativity via layout guidance

Computer Vision Center, Barcelona

📅 March, 2024 – Present

GRAIL: Graph reasoning for automatic interpretation and learning (PID2021-126808OB)

Computer Vision Center, Barcelona

📅 June, 2022– Oct, 2022

Low-cost real-time vision-based Industry 4.0 solution for metrological inspection of the automobile products. (Patent no: 202031050404)

Indian Institute of Technology Kharagpur

📅 Feb 2020–May 2020

TECHNICAL SKILLS

C/C++, Python, R, Matlab

MySQL, SQLite.

numpy, pandas, matplotlib, seaborn, scikit-learn

opencv, folium, tkinter

Keras, Tensorflow 2.0, Pytorch

HTML, CSS, Bootstrap

Linux

git/Github

ACHIEVEMENTS

- Best student paper award, ICDAR 2024
- FI-SDUR and FI-Joan Oró Ph.D. fellowship
- 1st in INDIAai Lab2Market organised by NASSCOM.
- 1st in Researchathon 2021 in healthcare track organised by United Nations Environment Program, and Harvard Kennedy School.
- Winner of Google Build for Digital India, 2020

CERTIFICATIONS

- Machine Learning by Stanford University.
- Data Science Specialization with Python by IBM.
- Data Science Specialization with R by Johns Hopkins University.
- Programming, Data Structure, Algorithms using Python by IIT Madras.